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Faculty of Science
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Yaoundé, Cameroon

March 25, 2026

Dr. Anne B. McCoy
Editor-in-Chief
The Journal of Physical Chemistry Letters
American Chemical Society

Dear Dr. McCoy,

We submit “Quantum-Coherent Spectral Engineering: Non-Markovian Dynamics in Photosynthetic Energy Transfer under Engineered Photon Baths” for consideration as a Letter in The Journal of Physical Chemistry Letters.

What we found: Spectral engineering of the incident photon bath extends excitonic coherence lifetimes by 20 % to 50 % and enhances excitonic energy transfer efficiency by 25 % through non-Markovian dynamics. Dual-band filtering at 750 nm and 820 nm targets vibronic resonances in the Fenna–Matthews–Olson complex, producing polaron-dressed states with reduced effective reorganization energy and suppressed pure dephasing.

Why it matters: This work establishes *spectral bath engineering* as a general principle for coherence control in open quantum systems. Unlike previous studies that treat environmental coupling as a constraint, we demonstrate that deliberate spectral shaping can *enhance* rather than suppress quantum effects. The mechanism—vibronic-resonance-assisted transport—is applicable beyond photosynthesis to any light-driven quantum process.

Why JPCL: Our findings bridge quantum dynamics theory with potential experimental realization via programmable light sources or filtered illumination in 2DES experiments. The concise 5-page format is ideal for presenting this focused physical insight without extensive materials discussion.

Key contributions:

1. Demonstration of 20 % to 50 % coherence lifetime extension via spectral filtering
2. Physical mechanism: vibronic resonance condition selectively suppresses pure dephasing
3. Coherence enhancement ($\eta = 0.20 \pm 0.04$) persists under physiological

disorder ($\sigma = 50 \text{ cm}^{-1}$) and temperature (285 K to 300 K)

4. 12-test validation suite confirms computational accuracy (HEOM benchmark: 1.8 % deviation)

This manuscript is a computational/theoretical study. All simulations use the Process Tensor Hierarchy of Pure States (PT-HOPS) method with Spectrally Bundled Dissipators (SBD), a numerically exact framework for non-Markovian quantum dynamics. Experimental validation via 2DES under filtered illumination is a promising direction for future work.

All authors have approved this manuscript. We declare no conflicts of interest.

Suggested reviewers:

- Gregory Scholes (Princeton) – gscholes@princeton.edu
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- Martin Plenio (Ulm) – martin.plenio@uni-ulm.de
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Excluded reviewers: None requested.

This manuscript is not under consideration elsewhere. Upon acceptance, we will transfer copyright to ACS.

Thank you for your consideration.

Sincerely,

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